



Airline Passengers: Satisfaction Recognition through Machine Learning

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Introduction and Goals

in the role of an airline company, the aim is to understand ...



- Is the survey useful for the company?
- Is Unsupervised Learning to identify satisfied and unsatisfied clients?
- Which is the best classification algorithm for our dataset?

Description of the dataset

The dataset contains 25,976 instances and 25 features that describe the results of a customer satisfaction study submitted to the passengers of an airline company.

The 25 variables can be divided into..

- Facts, such as Gender
- Opinions, such as Food and Drink
- The target feature: Satisfaction

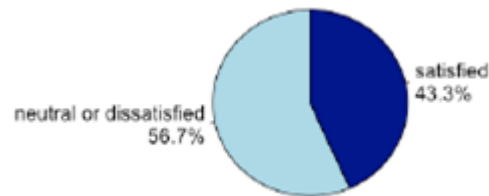


Figure 1: satisfaction pie chart



Pre-processing

- Missing values: Arrival.Delay.In.Minutes & evaluation scale issue
- Outliers in Departure.Delay.In.Minutes and Arrival.Delay.In.Minutes
- Graphical considerations: Class & Satisfaction

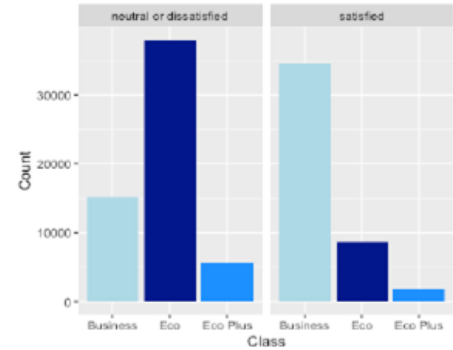


Figure 2: conditional distribution for "class"

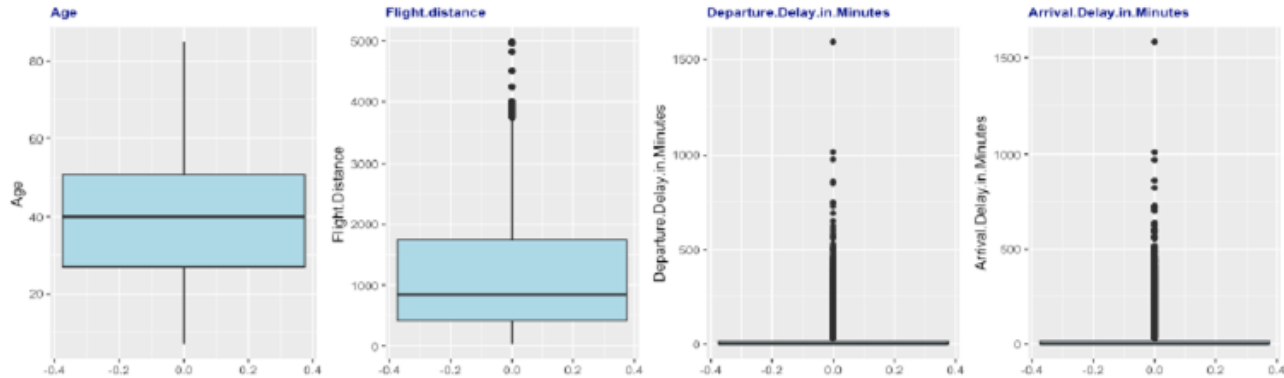


Figure 5: focus on boxplots which show outlier observations.



- Correlation analysis between quantitative variables
- Arrival.Delay.In.Minutes removed to avoid multicollinearity problem

Dbscan for the outlier analysis

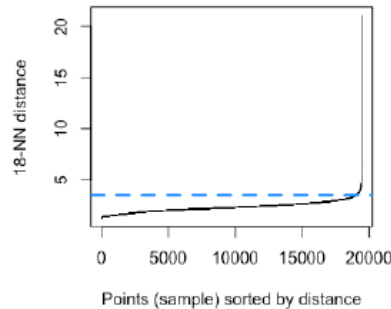


Figure 7: KNN graph

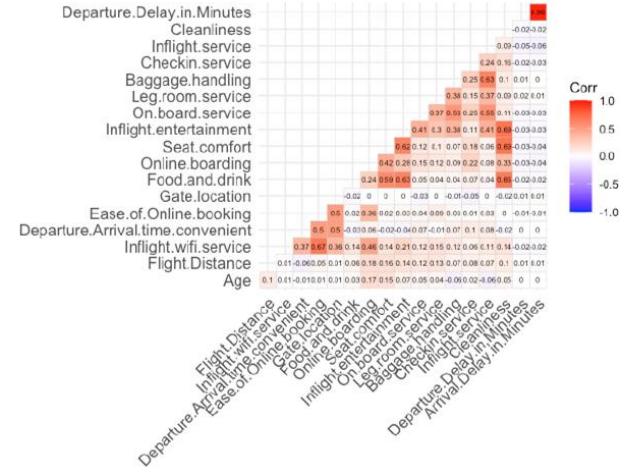


Figure 6: Corrplot between numerical features

hyperparameters: $\epsilon = 3.5$ & $\tau = 18$

- There's a correspondence between the noisy points and the outliers
- No changes in percentage of Satisfaction for the outliers which mean that they aren't important for the analysis
- Decision to delete these extreme values



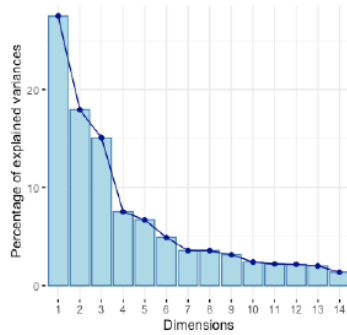


Figure 8: Scree plot graph for PCA

For the following clustering algorithm it has been decided to consider only rating features and to reduce their dimensionality through **PCA**

Its utility has been confirmed by some statistical tests like Bartlett's test of sphericity and KMO tests.

It has been possible to name the 7 components of the PCA in order to be able to explain the next results.

K-means

- The choice of K based on:
 - within and total distance ratio → **K=4**
 - silhouettes → **K=2**
- Are the clusters obtained significant?

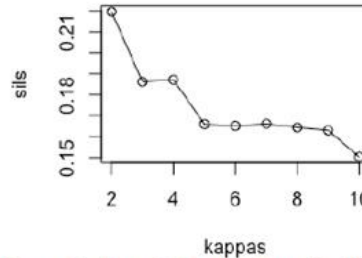


Figure 11: K maximisation through silhouette

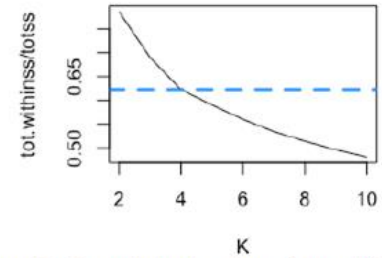


Figure 10: K maximisation through tot.withinss/totss

Hierarchical clustering

- The choice of the K through silhouette and of the distance measure
- How are the clusters obtained?
- Can the class labels be used as Cluster identifiers?

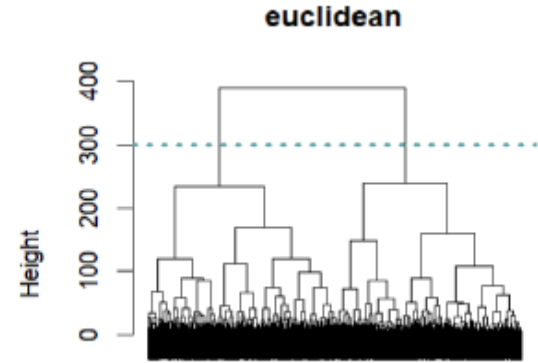


Figure 14: dendrogram



Figure 15: Pie chart for cluster 1

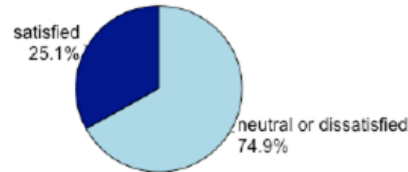


Figure 16: Pie chart for cluster 2



Figure 17: Scatterplot to compare the clusters (rating variables means)



Random Forest

hyperparameter: Size of the forest = 80

Results obtained in training and test set:

Classification accuracy: 94.49%

Sensitivity: 96.41%

Specificity: 92.06%

Time taken: 8.42 sec elapsed

Test accuracy: 94.78%

Sensitivity : 96.36%

Specificity : 92.71%

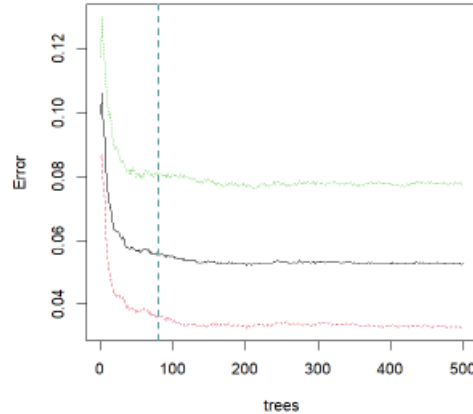


Figure 18: Relation between errors and number of trees

- The choice of the hyperparameter
- How does this algorithm perform in the dataset?
- Which are the least and most important features?
- Are there significant conclusions about rating variables?

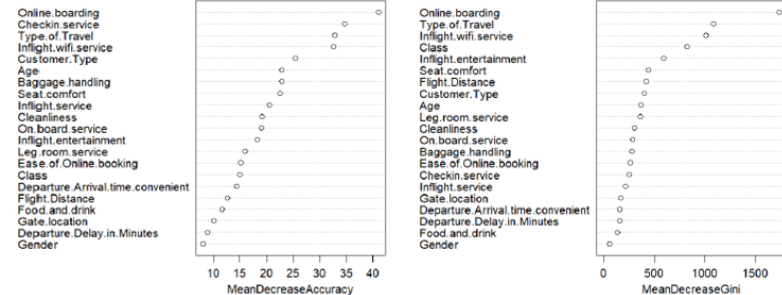


Figure 19: Variable importance plot for random forest with 80 trees



K-Nearest Neighbour

hyperparameter: k=9

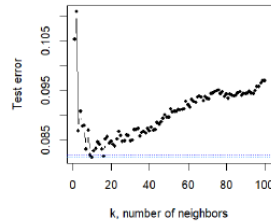


Figure 20: Test error rate VS number of neighbours

- Minimum point reached: k=10
- Binary context: KNN-9
- Euclidean Distance

Test accuracy: 91.81 %
Sensitivity: 87.81%
Specificity: 94.86%
Time taken: 4.84 sec elapsed

Adaboost with tree classifiers

hyperparameter: tree_depth=11 & n_rounds= 90

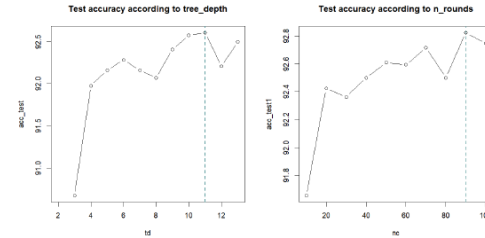


Figure 21: Hyperparameters choice for Adaboost algorithm

- Brief introduction to the algorithm
- The choice of the hyperparameter
- How does Adaboost algorithm perform in the dataset?

Training accuracy: 100%
Sensitivity : 100%
Specificity : 100%
Time taken: 46.44 sec elapsed

Test accuracy: 92.82%
Sensitivity: 90.94%
Specificity: 94.27%



Support Vector Machine

hyperparameter: cost= 0.5

- A linear kernel has been used
- The optimization of the hyperparameter
- How does SVM algorithm perform in the dataset?
- How could this results be improved?

Training accuracy: 88.64%

Sensitivity: 86.69%

Specificity: 90.17%

Time taken: 32.3 sec elapsed

Test accuracy : 89.16%

Sensitivity : 87.17

Specificity : 90.68%

Classification results

- Comparison of the classification algorithms
- Which is the best in terms of the indicators?
- What does the ROC curve show?

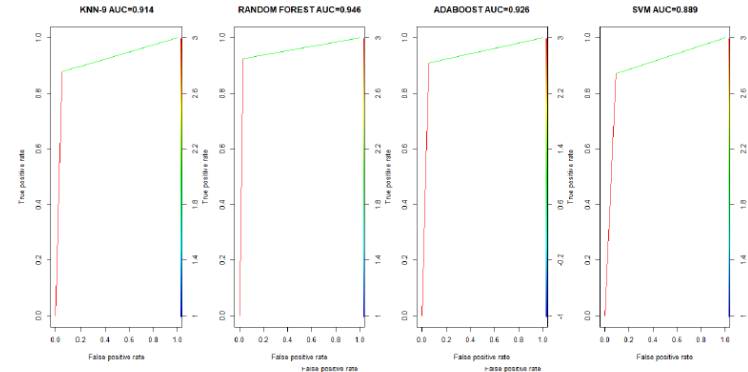


Figure 22: ROC Curve for the algorithm used in our analysis.



Conclusions

- The Random Forest algorithm is considered the best for classification
- This analysis shows non-trivial conclusions about the most influential variables for the response
- Possible suggestions to the airline company

